Experiment Optimization Plan

Sweeps, Ablations, and Open Decisions for the EB-JEPA EEG Track

EB-JEPA EEG track

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Abstract

A ranked catalogue of every sweep, ablation, and methodological fix still open on this project, organized by expected information gained per GPU-hour rather than by category. Cross-referenced against `reports/DECISION_LOG.md`'s open decisions (D1–D6) and `reports/EXPERIMENTS.md`'s "considered but not run" list, so this plan extends existing open questions instead of duplicating them.

1 Statistical rigor (cheapest, currently missing entirely)

1. **Multi-seed training.** Every encoder/loss config run so far used a single seed (`-single` in the launcher, for speed). The launcher already supports `-sweep <name>` with 3 seeds (1/1000/10000) for free. Re-running the top 2–3 configs with 3 seeds would tell us whether ConvNet's lead over LaBraM is real or noise.
2. **Bootstrap / clustered confidence intervals** on probe metrics. Windows from the same recording are not i.i.d.; a naive CI understates uncertainty. Nobody has computed this yet.
3. **Probe-only seed variance.** Re-fitting the MLP probe with 5 different `random_state` values costs nothing (no retraining) and separates probe variance from encoder variance.

2 Loss / objective coefficient sweeps

- **VICReg**: now fixed to the paper's $(\lambda, \mu, \nu) = (25, 25, 1)$, but worth sweeping around it – e.g. $(15, 15, 1)$, $(40, 40, 1)$ – per architecture, since the bug's effect was architecture-dependent (catastrophic for EEGPT, mild for Conv).
- **SIGReg**: `num_slices` $\in \{64, 128, 256, 512\}$, `lmbd` $\in \{1, 10, 50\}$.
- **Predictive JEPA**: EMA decay `ema` $\in \{0.99, 0.996, 0.999\}$ (too fast $\Rightarrow$ target chases online, defeating the point; too slow $\Rightarrow$ barely updates); prediction horizon (currently predicts every future frame one-by-one from frame 0 – could instead predict only frame $+1$, or frame $+k$ for varying k); `pred_std_coeff`/`pred_cov_coeff`.

3 Augmentation ablations (cheap, high diagnostic value)

- **Leave-one-out**: disable each of the 4 augmentation steps (scale jitter, noise, channel-drop, time-mask) individually and retrain, to isolate which one actually drives the gain. `EXPERIMENTS.md` explicitly flags this as never separated for the corruption variant either.
- **EEGPT-specific**: disable channel-drop only for EEGPT (**D1** in the decision log – channel-drop conflicts with EEGPT's channel-pooling bottleneck; jobs 75752/75806 test this).
- **Exact-corruption coefficients** (`aug_mask_frac`, `aug_outlier_frac`, `aug_outlier_scale`): only the spectral-loss coefficients (0.05/0.1/0.3) were swept; the corruption-specific knobs themselves were not.

4 Architecture hyperparameter sweeps

- Conv: `hidden` / `depth` / `kernel_size`.
- Transformers: `embed_dim`, `tr_depth`, `n_heads`, `patch_len` (1 s patches were never compared against e.g. 0.5 s or 2 s).

- **Capacity-matched comparison (D6, open).** Conv (∼0.4M params) is currently compared to LaBraM/EEGPT/BIO at whatever size they happen to be – "architecture wins" and "more capacity wins" are confounded. Scaling every encoder to an equal parameter count before comparing is the single biggest methodological gap in the architecture comparison.

5 Training schedule

- **Epoch count**: everything so far is 20 epochs. LaBraM trained ∼200 epochs on a much larger corpus – 20 may simply be too few for transformers to converge, independent of architecture quality.
- **LR schedule**: currently flat AdamW, no warmup/decay – a cosine schedule could matter more for transformers than for Conv.
- **Batch size**: VICReg's covariance term needs enough samples per batch to estimate decorrelation meaningfully; a larger batch could shift the variance/covariance balance.

6 Data scale

- `subset_frac` **scaling curve**: 5% → 25% → 50% → 100%, plotted against probe BACC – reveals whether any encoder is data-starved (transformers usually need more data than Conv before showing their advantage).
- **Multi-corpus** (TUAB+TUEV+TUSZ+TUEP, tried once on Conv only by tvasnier) – never tried on the transformer encoders, where extra data might matter more.

7 Probe / eval protocol

- **LogReg vs. MLP probe (D3, open)**: only MLP has been used so far. LogReg is the literature standard, more directly comparable to SimCLR/MoCo-style papers, and removes the probe-capacity confound.
- `n_windows` **sweep** (8/16/32/all) for per-recording pooling; now that per-window scoring exists, compare both protocols at multiple N.
- **Re-score every existing checkpoint under both protocols** (per-window now implemented but not yet run) for a complete dual table.

8 Cross-component combinations (each tested in isolation so far)

- **Spectral loss + corrected VICReg (D4, open).** The best single result (0.836 per-recording) used the spectral auxiliary loss but with the *buggy* VICReg coefficients. Never retrained with the fix – open question whether spectral was adding independent signal or just compensating for the bug.
- **Corruption + SIGReg** (lightly tried) vs. **corruption + spectral** (tried, regressed) – the fuller cross product (augmentation × loss × anti-collapse mechanism) has not been run.
- **Predictive JEPA + SIGReg** anti-collapse instead of the current Hinge-std/covariance pair.

9 Architectural fixes worth isolating

- **BIOT with phase**: currently FFT-magnitude only (discards phase) – adding a complex/phase channel to the token embedding directly tests whether phase loss is the bottleneck.
- **EEGPT without the channel-pool bottleneck**: replace the channel-summarization attention with simple mean-pool across channels, to test whether the hierarchical design itself (not just channel-drop) is the problem.
- **LaBraM-style tokenizer warm-start (D2, open)**: pretrain the patch embedding with a quick reconstruction objective before SSL, closer to the real LaBraM recipe.

10 Extensions beyond TUAB

- **TUEV** (6-class, harder, imbalanced) – the original README's suggested follow-up, untouched.
- Other datasets already scoped in `suggestions/datasets/`: BCI-IV-2a, HMC, sleep-EDF, HGD – would test whether any encoder's advantage is TUAB-specific or general.

11 Ranked recommendation

By expected information gained per GPU-hour, in order:

1. Multi-seed runs on the existing best 2–3 configs (Section 1).
2. Capacity-matched architecture comparison (Section 4).
3. Augmentation leave-one-out (Section 3).
4. Epoch-count sweep for the transformer encoders (Section 5).
5. Spectral + fixed-VICReg retrain, i.e. D4 (Section 7).

These five directly resolve open questions already on record, rather than opening new ones – the highest-leverage use of remaining compute.